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Experimental and Numerical Investigation of Phase Change Material (PCM) Heatsink for Electronic Thermal Management Applications

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Abstract:

The increasing thermal requirements of electronic devices necessitate the use of effective passive cooling technologies. In present study, the thermal performance of phase change material (PCM)-integrated heat sinks is investigated using a combined experimental and numerical method. Acrylic heat sinks embedded with paraffin-based PCMs of different melting temperatures (42 °C and 55 °C) and thicknesses (60 mm and 120 mm) were tested under free convection conditions with heat inputs ranging from 5 W to 15 W. A three-dimensional transient numerical model based on the enthalpy–porosity method was also developed to simulate the phase change process and validate the experimental observations. The numerical model closely matches the experimental results, with about 13 % deviation confirming the reliability of the approach. Results demonstrate that PCM integration significantly reduces the rate of temperature rise and enhances thermal stability compared to conventional heat sinks without PCM. Increasing the PCM thickness from 60 mm to 120 mm extends the thermal regulation period by up to 50 minutes at 5 W and 35 minutes at 15 W. PCMs with higher melting temperatures extend thermal regulation, particularly at higher heat inputs, but lead to higher final operating temperatures. The findings emphasize that selecting suitable PCM melting temperature and thickness is crucial for effective thermal management in PCM based power electronic devices.

Keywords: Electronic devices, Power ratings, Heat sinks, Thermal regulation, Heat dissipation

1. Introduction

In the modern digital era, as electronic components and devices are increasingly integrated and quickly trimmed, the outcome is extraordinarily high-power dissipation rates and steadily rising

device temperatures. The operational efficiency of many electronic devices, such as electronic boards, cellphones, laptops, CPUs, and missile guidance systems, has decreased due to improper management of heat regulation [1]. According to survey results, almost 50% of electronic device malfunctions are caused by temperatures that are too high for the equipment to operate [2]. Electronic devices frequently undergo high-frequency power cycling, necessitating that their cooling systems effectively manage transient thermal profiles. This capability is critical for delaying temperature responses and minimizing temperature gradients within the device. Failure to adequately control these thermal dynamics can lead to thermal stresses, which may ultimately result in the degradation and failure of the electronic components over time. [3]. Substantial research is published on electronic device cooling using passive and active systems. Naphon et al. [4] investigated the thermal distribution characteristics of cold plate heat sinks incorporating various fin geometries, such as rectangular, circular, and conical fins. Their research focused on the influence of fin shape on the overall thermal resistance of the heat sink, revealing only a negligible reduction in heat-spreading resistance. Furthermore, the integration of passive heat transfer devices, including heat pipes and heat sinks that incorporate phase change materials (PCMs), has generated significant interest among researchers seeking to enhance thermal management in electronic systems, thereby improving the thermal performance of electronic components. The first report was proposed by Humphries et.al. [5] based on use of PCM for avionics control which opened the door for PCM based heat sinks. Later, several studies concentrated on the performance of PCM heat sinks for electronics thermal control, including the integration of PCMs in a heat sink [6-10]. Various enhancement techniques including experimental and numerical studies like different heat sink configurations, PCM quantity, PCM encapsulation heating rate and melting temperature [10] are used extensively in the present domain. Kandasamy et al. [11] conducted a study of the effect of a PCM-heat sink in enhancing cooling performance vs a non-PCM heat sink. This study examines the effect of heating power (2-6 W) from a horizontal, uniform heat source on a PCM heat sink. Wang et al. [12] conducted a parametric analysis of the performance of a finned heat sink filled with and without PCM.

This paper proposes an enhancement in the thermal performance of a heat sink system operating under free convection conditions. Critical determinants influencing system performance include the constant and pulsed temperature differentials between the input boundary temperature and the melting point of phase change materials (PCMs), aspect ratio, and various properties of the PCMs. Alawadhi and Amon's study [13] addressed both steady and unsteady power levels by performing experimental and numerical analyses of PCM-based heat sinks designed for portable

electronic devices. As per this research, the thermal control unit provides a dependable way to prevent overheating in portable electronics while simultaneously meeting ergonomic standards. Tauseef et al. [14] reported on the energy storage capabilities of thermal management in electronics utilizing the PCM RT-35HC paraffin. This study pointed out that using a PCM volume fraction of 0.8 significantly decreased the maximum base temperature by 22%, especially in a heat sink with copper foam at its center. Copper foams with a porosity of 97% performed better than those with a porosity of 95%.

The importance of choosing a suitable PCM melting temperature in a horizontally oriented heat sink configuration with heating sources ranging from 60 to 120 W was also investigated by Fan et al. [15]. Hexadecanol and eicosane were amongst the PCMs under investigation. Findings of this study suggest that using a PCM with higher melting point temperature led to a longer thermal regulation period. In contrast, PCMs with lower melting temperatures were better at quickly lowering the maximum temperature in a short amount of time.

Use of phase change materials for thermal regulation is not new. Integration of phase change materials for applications such as temperature regulation in EV batteries [16], free space cooling [17], maintaining temperature in cold rooms [18] have proved to promote energy efficiency. Despite extensive research into PCM-based heat sinks for electronic cooling, following gaps are recognized.

Gaps in Literature

- PCM melting temperature has significant impact on thermal performance [15, 19]. However, limited comparative experimental studies exist under low-to-moderate heat loads (5–15 W), which are characteristic of consumer and portable electronic devices.
- A significant portion of the literature focus on high heat flux conditions or short-duration thermal shocks [5, 14], which restricts their applicability to real-world passive cooling situations.
- Many literatures highlighted performance enhancement using complex structures such as topology-optimized inserts, metal foams, and finned or metallic heat sinks [6,8,14], which increase fabrication cost and design complexity.
- Existing studies have largely overlooked low-cost and simplified heat sink configurations, especially those that allow direct observation of PCM melting behavior [4, 19].

- Moreover, limited efforts have been made to integrate experimental and 3D numerical analyses for the simultaneous investigation of the coupled effects of PCM melting temperature and thickness under identical operating conditions.

Key Contributions

Through a combined experimental and numerical investigation, the proposed study attempts to make the following contributions:

- Investigation under realistic operating conditions:

The study systematically evaluates the thermal performance of PCM-filled heat sinks under low-to-moderate power inputs (5–15 W), and proves applicability of practical electronic devices, unlike many prior studies focused on high heat flux or transient thermal shocks.

- Experimental correlation of thermal performance:

An experimental correlation is established for PCM-based heat sinks operating under free convection, demonstrating the role of latent heat storage in justifying abrupt temperature rise and enhancing thermal stability.

- Controlled comparative analysis of PCM melting temperatures:

A direct experimental comparison between two paraffin-based PCMs (OM42 and OM55) is conducted systematically under identical geometrical and operating conditions. The results clearly quantify the trade-off between lower peak temperature reduction (lower melting PCM) and extended thermal regulation duration (higher melting PCM), providing insights beyond general trends reported in literature [15, 19].

- Study of the effect of PCM thickness on thermal regulation using a simplified configuration:

The influence of PCM thickness on thermal regulation and heat input is analyzed using geometrically scaled, low-cost acrylic heat sinks, offering practical insights without relying on complex enhancement techniques such as fins or metal foams.

- Integration of experimental results with 3D numerical modelling:

The present study combines direct visualization of PCM melting behaviour with a three-dimensional transient numerical model based on the enthalpy–porosity formulation showing good agreement and improving physical understanding and model validation.

- Design-oriented outcomes for practical applications:

The work provides application-specific design guidelines for selecting PCM melting temperature and thickness based on power input (5–15 W) and required thermal regulation duration, thereby enhancing the applicability of PCM-based passive cooling systems.

The efficiency of thermal management systems is governed by the geometry of the heat sink, the thermo-physical characteristics of the materials used, and the current operating conditions. In the present context, our research work emphasizes utilization of PCM for electrical thermal management application.

2. Methodology:

To estimate the potential for phase change material based thermal management of heat sinks, a combined numerical and experimental procedure was built up. The main objective is to examine the effect of thermal behavior of heat sink by optimizing the parameters like multiple-PCM design, PCM thickness, melting temperature, and heat source intensity. The steps followed in the investigation are shown in Figure 1.



Figure 1. Steps showing the execution of the study

Each of these steps is elaborately presented in the following sub-sections.

2.1 Selection of PCM

One of the main considerations in selection of the appropriate type of PCM is the expected operating temperature range of the electronic device. A safe operating temperature for these devices is generally below 70°C. Paraffins are one of the most easily available and low cost PCMs available. The phase change materials (PCMs) used in the present study OM42 and OM55 are commercially available paraffin-based organic PCMs supplied by Rubitherm GmbH. These

materials are widely used in thermal energy storage and electronic cooling applications due to their stable thermophysical properties, chemical inertness, and reliable phase change behavior. Both OM42 and OM55 are part of Rubitherm's RT/OM series of paraffin-based PCMs and are specifically designed for applications requiring controlled temperature regulation within the range of 40–60 °C. The manufacturer provides detailed material specifications, including melting temperature range, latent heat, density, and thermal conductivity, which were used as the basis for experimental design and numerical modelling in this study.

Organic paraffin-based PCMs (OM 42 and OM 55) offer stable melting behaviour, and excellent compatibility with electronic heat sinks, making them highly suitable for electronic thermal management. Table 1 displays properties of the considered PCMs and compares them to that of salt hydrates and Fatty acids which are used as potential phase change materials.

It may be noted from Table 1 that although salt hydrates provide higher latent heat and thermal conductivity, their practical application is limited by phase segregation, supercooling, and corrosiveness. Fatty acids exhibit good thermal reliability and comparable latent heat; however, their higher cost and potential material compatibility issues restrict large-scale use. Therefore, organic PCMs were selected in the present study to ensure reliable, safe, and repeatable thermal regulation under varying heat loads.

Table 1. Properties of different PCMs used for the experimental work

Property	Organic PCM (OM 42)	Organic PCM (OM 55)	Salt Hydrate PCM	Fatty Acid PCM
PCM category	Organic	Organic	Inorganic	Organic
Typical examples	Paraffin-based	Paraffin-based	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	Capric acid, Lauric acid
Melting temperature range (°C)	41–43	54–56	30–60	30–60
Peak melting temperature (°C)	42	55	35–55	35–55
Latent heat of fusion (kJ kg ⁻¹)	190–220	200–230	200–280	180–210
Density (solid phase) (kg m ⁻³)	~860	~870	1400–1600	850–940
Density (liquid phase) (kg m ⁻³)	~770	~770	1300–1500	820–900
Thermal conductivity (solid) (W m ⁻¹ K ⁻¹)	0.20–0.25	0.20–0.25	0.5–1.0	0.2–0.3

Thermal conductivity (liquid) (W m ⁻¹ K ⁻¹)	0.15–0.18	0.15–0.18	0.4–0.6	0.15–0.25
Specific heat capacity (kJ kg ⁻¹ K ⁻¹)	~2.0	~2.1	1.4–1.8	1.9–2.2
Volume expansion (%)	8–12	8–12	5–10	6–10
Phase segregation	None	None	High	None
Supercooling tendency	None	None	Moderate–High	Low
Corrosiveness	Non-corrosive	Non-corrosive	Corrosive	Non-corrosive
Thermal cycling stability	Excellent	Excellent	Moderate	Good
Compatibility with containers	Excellent	Excellent	Limited	Good
Cost	Low–Moderate	Low–Moderate	Low	Moderate
Suitability for electronics cooling	Excellent	Excellent	Limited	Moderate

In the present set of experiments commercial PCM OM 42, and OM 55 as listed in Table 1 were selected as their melting point temperature matched that of the application in question. These materials have similar latent heat capacity (~200 kJ/kg), density in liquid phase (860 kg/m³), density in solid phase (~900 kg/m³), and thermal conductivity (0.2 W/m K).

To validate and establish confidence in the experimental results, a computational simulation of the suggested setup was performed to numerically estimate the heat generation and associated temperature rise. The details of the numerical model are provided in the next section.

2.2. Numerical modelling to estimate PCM temperature

Despite the fact that multiple research studies have been conducted using a 2D domain, it has been found that the phase change phenomena cannot be detected critically in all directions. As a result, a 3D model of the PCM heat sink is created using the ANSYS Workbench design modeler. In order to understand the effect of PCM mass (thickness) in a heat sink, a rectangular design with variation in the length was explored. Two geometries of heat sink viz., Heat sink A - 60 x 60 x 60 mm (as shown in Figure 2) and Heat Sink B - 60 x 60 x 120 mm made of acrylic material was conceptualized.

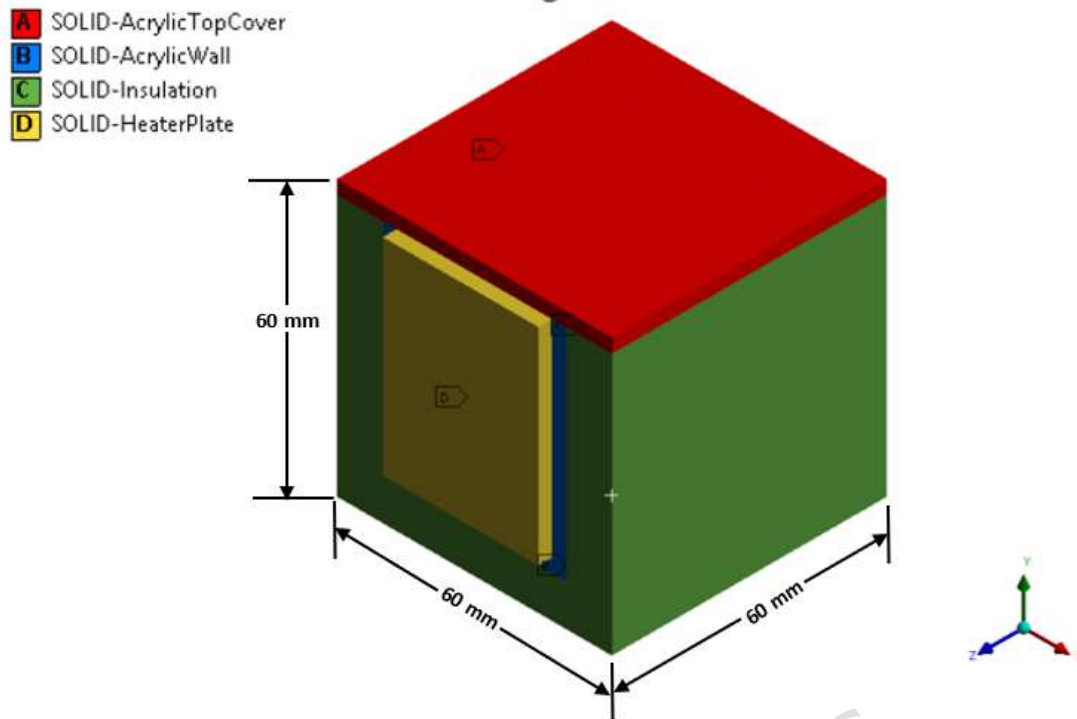


Figure 2. 3D model of Heat sink A

A constant heat flux is provided by a 3 mm thick heater plate on one side. To aid the heat transfer process, a 3mm aluminum plate is fixed next to the heater plate replacing the acrylic wall. The upper portion is covered with clear acrylic plate to observe the PCM melting profile while the remaining 4 sides are insulated with polystyrene. The major dimensions and inputs for the simulation are listed in Table 2 and Table 3.

Table 2. Heat sink A: Dimensions for the set up and numerical simulation

Parameters	Specifications
Overall dimension	60mm x 60mm x 60mm
Heater plate dimension	45mm x 34mm x 3mm
Aluminium plate dimension	45mm x 34mm x 3mm
PCM volume	45mm x 34mm x 44mm
Insulation thickness	10mm on 3 sides and 9mm on bottom side
Acrylic wall thickness	3 mm on 3 sides including top and bottom
Heat input through heater plate	5W

Table 3. Properties of solid material

Part Description	Material	Density ρ , (kg/m ³)	Specific Heat, C_p (J/kgK)	Thermal conductivity, k (W/mK)
Heater plate	Aluminium	2719	871	202.4
Insulation	Polystyrene	30	1400	0.03
Acrylic Wall	Acrylic (PMMA)	1180	1470	0.2

2.2.1 Simulation settings

The computational model employs the finite volume method with enthalpy-porosity formulation to solve the governing equations subject to initial and boundary conditions. In this technique the liquid fraction of the cell volume is calculated where the domain is considered to be a porous media with porosity ranging from 0 in the solid phase and 1 in the liquid phase while the mushy zone in between 0 to 1.

The following assumptions are made while developing the numerical model:

1. The PCM slab (control volume) is homogenous and isotropic.
2. PCM in its liquid phase is considered as a newtonian and incompressible fluid resulting in laminar flow inside the PCM slab.
3. The change in volume of PCM due to phase change is neglected.
4. PCM thermophysical properties only depend on the solid or liquid phase and are temperature independent.
5. During the phase change process of PCM the solid-liquid interface is defined as a region called the mushy zone where the PCM is assumed to a mixture of both its liquid and solid phase.
6. The PCM slab is insulated, hence outer wall barring the heat plate side, is taken to be adiabatic.
7. Transient energy model is used while neglecting the effect of radiation and viscous dissipation.
8. No heat generation inside the PCM

The governing equations solved to predict the transient distribution of the temperature and the liquid fraction of the PCM are as follows [22]:

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{U}) = 0$$

where ρ and $\vec{U}$ are the density and velocity respectively.

Momentum equation:

$$\frac{\partial(\rho \vec{U})}{\partial t} + \nabla \cdot (\rho \vec{U} \vec{U}) = -\nabla P + \rho \vec{g} + \nabla \vec{\tau} + \vec{F}$$

where P is the static pressure, $\vec{\tau}$ represents the stress tensor, $\rho \vec{g}$ denotes the gravitational force and $\vec{F}$ is the external body force.

Energy equation:

$$\frac{\partial(\rho H)}{\partial t} + \nabla \cdot (\rho \vec{U} H) = \nabla \cdot (k \nabla T) + S$$

where H and k is the enthalpy and thermal conductivity of the PCM, T represents the temperature and S is the volumetric heat source. The total enthalpy H of the PCM is represented as

$$H = h + \Delta H$$

where h and ΔH are the sensible and the latent heat of the PCM respectively.

The sensible heat h is given as

$$h = h_{ref} + \int_{T_{ref}}^T C_p dT$$

where h_{ref} and T_{ref} are the reference enthalpy and temperature respectively while C_p is the specific heat at constant pressure.

Also, the latent heat of PCM for a given liquid fraction is expressed as

$$\Delta H = \beta L$$

where β is the liquid fraction and L is the latent heat of the melting. The liquid fraction β is defined as

$$\begin{aligned} \beta &= 0 & T &< T_{solidus} \\ \beta &= \frac{T - T_{solidus}}{T_{liquidus} - T_{solidus}} & T_{solidus} &< T < T_{liquidus} \\ \beta &= 0 & T &> T_{liquidus} \end{aligned}$$

2.2.2 Grid independence study

A mesh independence study is crucial in CFD to ensure that the discretization of the computational domain does not influence its numerical outcomes. The accuracy of the solution depends on the grid resolution since the governing equations are solved on a finite mesh. In this work, simulations

are performed using three progressively refined mesh sizes and the key PCM parameters are compared to evaluate their sensitivity to grid size. Figure 3 displays these three mesh resolutions generated for Heat sink A, denominated as coarse (Figure 3a), medium (Figure 3b) and fine (Figure 3c) meshes with 1926, 68620 and 516400 elements, respectively.

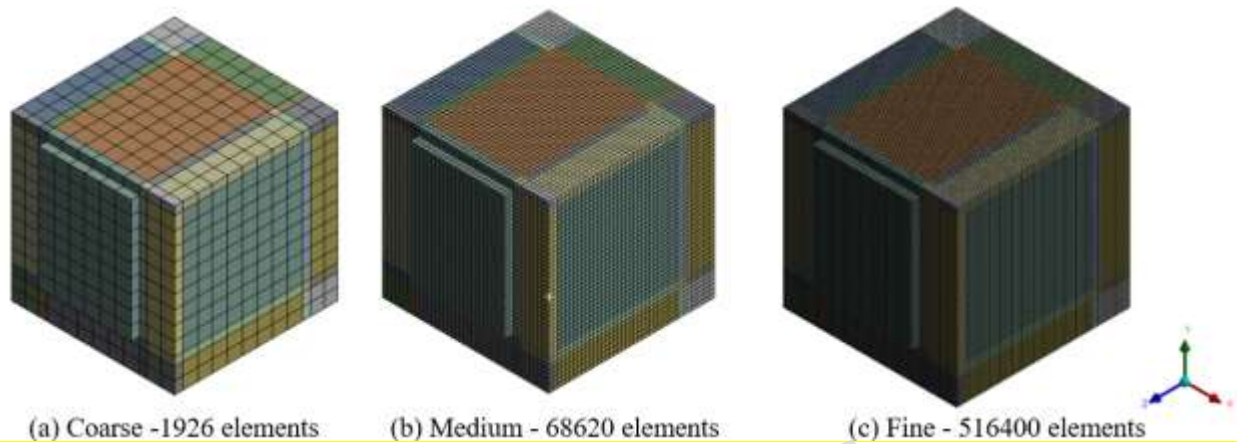


Figure 3. Coarse, medium and fine computational meshes for Heat sink A

The mesh independence was evaluated by comparing the temperature profile and liquid fraction of PCM in each of these three mesh sizes, which is represented graphically in Figure 4a and 4b.

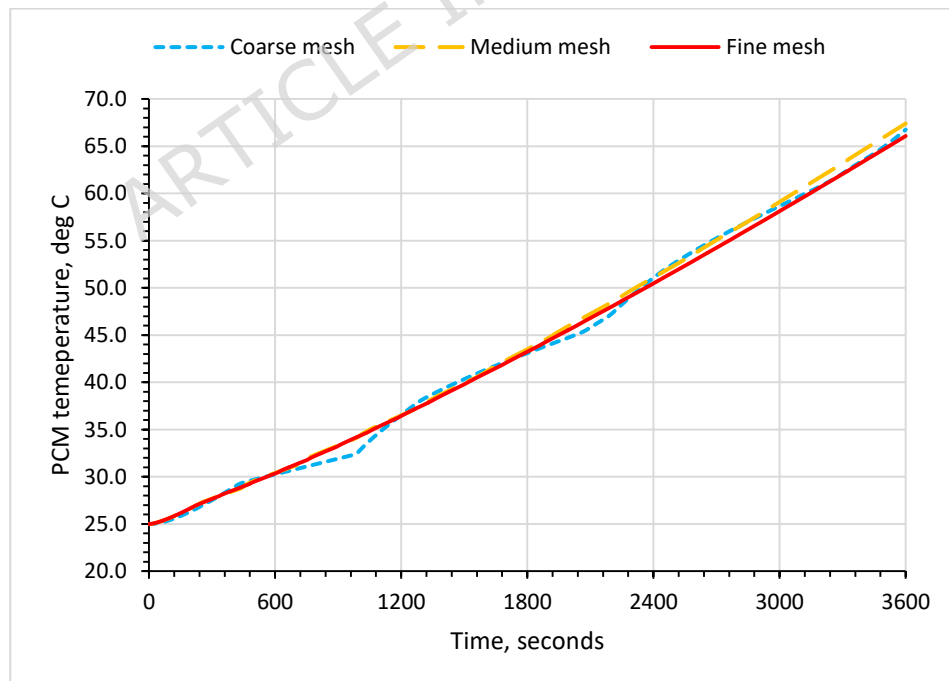


Figure 4a. Mesh size evaluation for PCM temperature in heat sink A

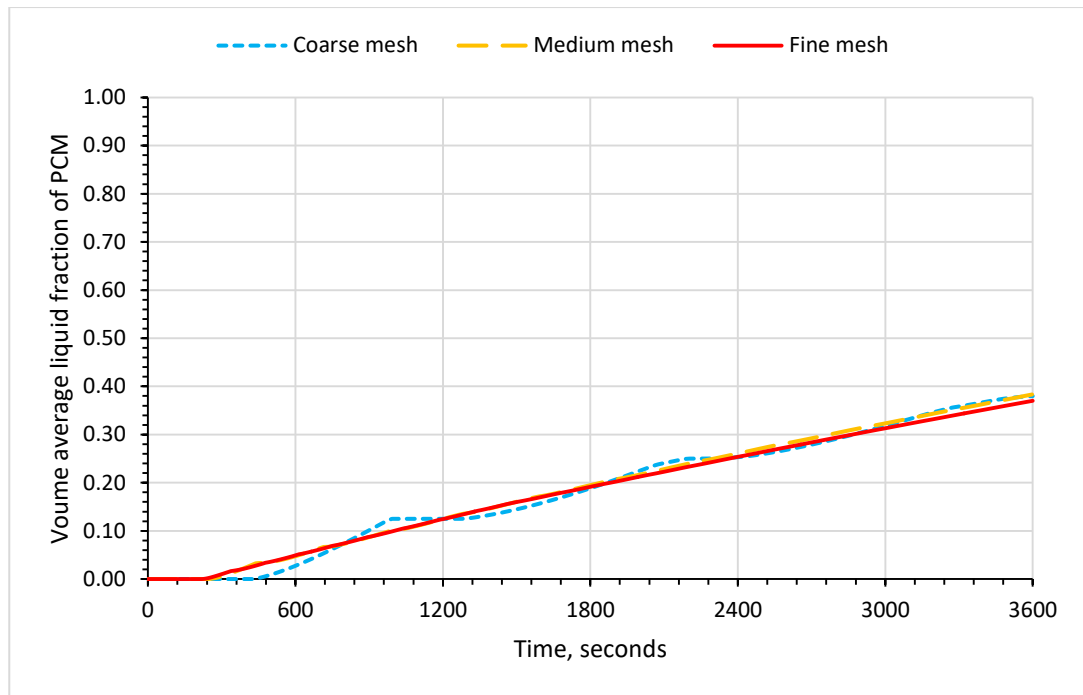


Figure 4b. Mesh size evaluation for PCM liquid fraction in heat sink A

It is noted that in the study for the three meshes, the PCM temperature variation is restricted to a maximum of 1.9%, while the liquid fraction varies by a maximum of 2.6 % at the end of 1 hour simulation, as evident from Table 4. Observing the smooth progression of properties under the fine mesh, this denser grid is further taken up for simulation to ensure accuracy in the results.

Table 4. Parameter comparison for grid evaluation in heat sink A

Mesh type	No. of elements	PCM temperature, °C	Liquid fraction
Coarse	1926	66.7	0.38
Medium	68620	67.4	0.38
Fine	516400	66.1	0.37

2.3. Experimental Set up of PCM Heat Sink:

The setup of PCM heat sink is developed to reduce overheating, avoid higher heat generation, and also provide uniform cooling to the commercial electronics devices which helps to improve lifespan, in addition improving performance and reliability of electronic devices. In our experimental study, two heat sinks with different dimensions are designed. Fig. 5 shows PCM heat sinks, designated as A and B having exterior dimensions of 60 x 60 x 60 mm, 120 x 60 x 60 mm, respectively, and are constructed of acrylic. To monitor PCM melting profile the upper portion is of clear acrylic plate. An aluminum plate with a thickness of 3 mm is used to divide in two cavities.

To heat the PCM, a 3 mm thick aluminum plate has been mounted to one side of the container and thermally insulated with polystyrene. The temperature is measured using PTC thermistor (Positive Thermal Coefficient). The aluminum plate temperature and the PCM temperature distribution across the container are measured using thermistors that are fixed within the container. It is observed that the thermocouple T1 is positioned in the center of the aluminum plate's inner surface to monitor the heater's temperature. The thermocouples from T2 to T5 are spaced 20 mm apart and are used to measure the PCM's temperature along the direction of heat flow. To measure the outside temperature, a thermocouple T6 is left outdoors. The temperature readings are taken using a data logger at every 60-second interval. In order to monitor the PCM's melting profile, a digital camera was utilized to take pictures of the heat sink every 15 minutes. In this study, PCM OM42 and OM55 with melting temperatures of 42°C and 55 °C, respectively, are used to analyze the impact of melting temperature on heat sink performance. First, the PCM was melted in order to fill the PCM container. After that, the container was filled with the melted PCM. It is assured that the PCM is acquired at an initial temperature comparable to the ambient value before commencing each experiment.

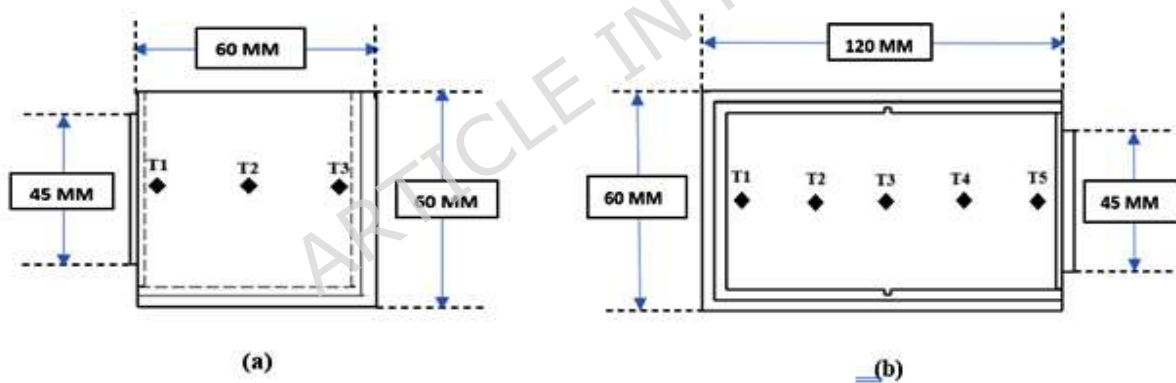


Figure 5. Position of temperature sensors and dimensions of heat sinks, (a) A, (b) B

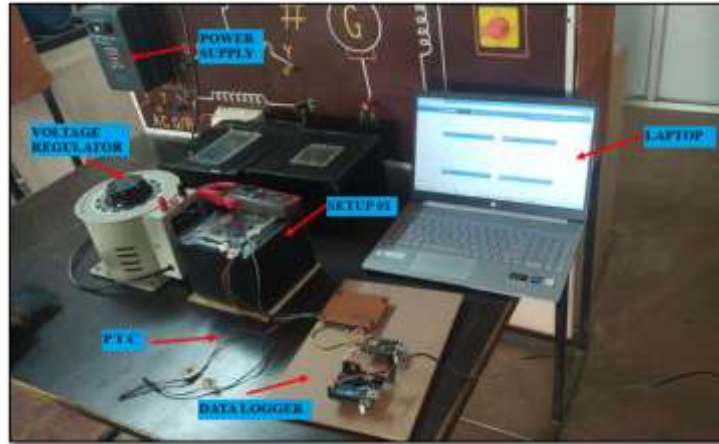


Figure 6. Experimental setup of PCM heat sink

It is mentioned that heat sink A with dimensions 60 x 60 x 60 mm is used for study and actual set up is shown in Fig. 6. The heat sink contains the PCM and is connected to a plate heater capable of generating up to 160 W. The temperature of the plate heater and the heat sink is measured at regular intervals of 60 seconds using PTC thermistors (PTC- Positive Thermal Coefficient). Data collection continues until the PCM has completely melted or up to reached to the steady state condition of a PCM heat sink.

2.4. Experimental Study

The study includes the effect of thermal behavior of heat sink by optimizing the parameters like multiple-PCM design, PCM thickness, melting temperature, and heat source intensity etc. Heat sink A is used for all design of experiments for duration 60 minutes. Table 5 shows the details of parameters varied during study.

Table 5: Parametric variation explored in present experimental study

Sr. No.	Parameter	Heat Sink	PCM Used	Heat Power Rating (W)	Duration (Min)
1	Effect of PCM on heat sink (with and without PCM)	Heat Sink A with dimension	OM 55	5,10,15	60
2	Effect of power rating	Heat Sink A with dimension	OM 55	5,10,15	60
3	Effect of melting temperature	Heat Sink A	OM 55 and OM 42	5,10,15	60

4	Effect of PCM thickness	Heat Sink A and B	OM 55	5	60
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3. Results and Discussion

This section details the results from the numerical simulation and experimental observations.

3.1. Numerical Simulations

The numerical simulation is carried out using ANSYS Fluent with the whole domain set to initial temperature of 298 K. All the outer walls being thermally insulated are at adiabatic conditions except the side where the heater plate is mounted is taken to be a constant heat source with heat supply of 5 W. The pressure-implicit with the splitting of operators (PISO) which was suitable to the transient simulation method was used in this study. PISO accurately couples pressure and velocity fields during transient flow to resolve buoyancy-driven motion. Natural convection arises in the model under gravity-enabled conditions, coupled with temperature gradient. Thus, the application of the PISO algorithm ensures improved convergence and stability for transient natural convection during phase change. For getting the more precise solutions second order upwind form of spatial discretization was used to solve the momentum and energy equations, while the transient approach used the second order implicit formulation. For a time step of 1 second the number of iterations is set as 20 per time step. This value was finalized based on preliminary runs, where the solution consistently converged in less than 20 iterations. This ensured efficient computation without compromising on the solution accuracy. Convergence is obtained when the residual of the energy, momentum and continuity equations are reduced to less than 10^{-6} , 10^{-3} and 10^{-3} respectively. The results of the simulation were recorded at regular intervals of 10 minutes for a duration on 60 minutes (1 hour). The results are represented as contours of liquid fraction and temperature of the PCM.

The influence of the PCM slab thickness on the heat transfer enhancement during charging of PCM is numerically investigated as explained in Section 2.2. Contours of liquid fraction and temperature of PCM are plotted at every time interval of 10 minutes of phase change from solid to liquid for OM55 with heat supply of 5W for Heat sink A and B.

Figures 7 and 8 represents the variation of liquid fraction with time for heat sink A respectively, while Figure 9 and 10 depicts the case of heat sink B. The shape and motion of melting interface with increase in time is represented through liquid fraction plots where the red color represents

the condition when material is completely liquid ($\beta=1$) and blue is indicative of its solid state of existence ($\beta=0$). The temperature plots demonstrate the heat input provided by the heater plate that penetrates through the aluminum plate to reach the PCM where the heat is absorbed and the latent heat of the PCM is utilized initiating the melting process in the PCM. The observed melting of PCM initially on both sides of the enclosure (as seen in the liquid fraction and temperature contour plots) is primarily governed by heat conduction paths and natural convection effects within the PCM domain. At the start of heating, heat is supplied through the aluminum plate and spreads into the PCM. The PCM melts first near the sides due to combined heat transfer effects. Heat spreads laterally through the aluminum plate and walls, creating higher temperatures near the edges. Faster thermal gradient development at the side walls initiates early melting in these regions. As melting begins, natural convection enhances heat transfer along the sides, further accelerating melting there. In contrast, the central region melts later since it relies mainly on slower conductive heat transfer from surrounding liquid PCM.

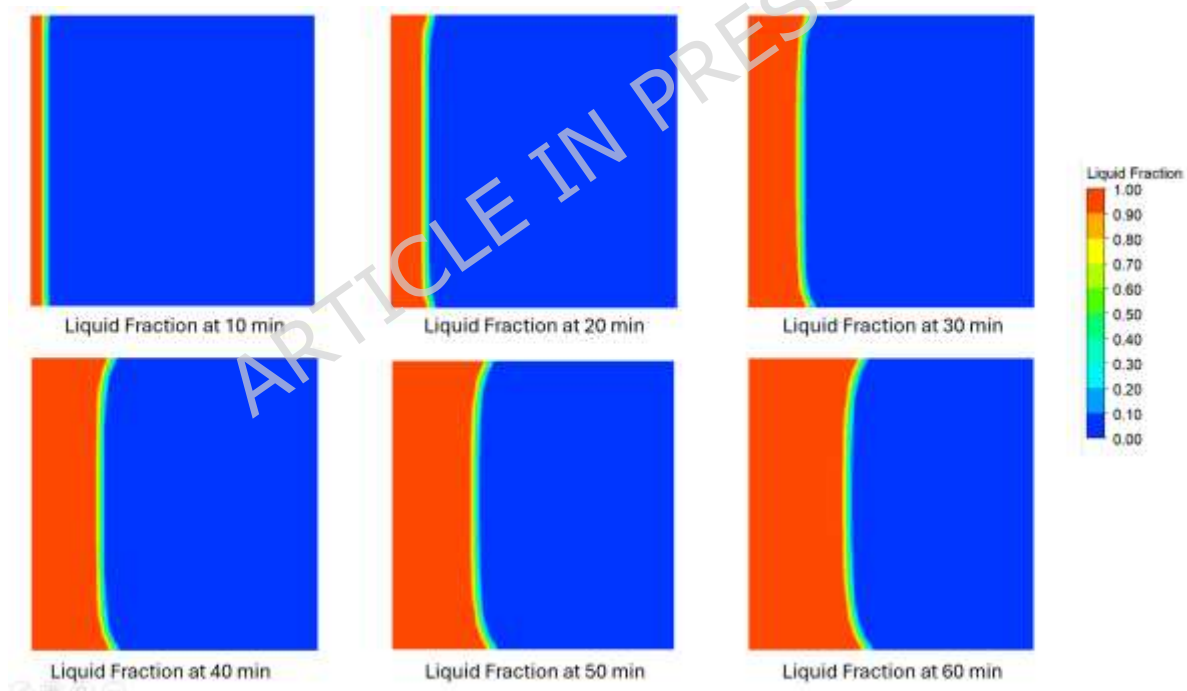


Figure 7. Variation of PCM liquid fraction over 1 h time duration for heater input of 5 W for Heat sink A using OM55

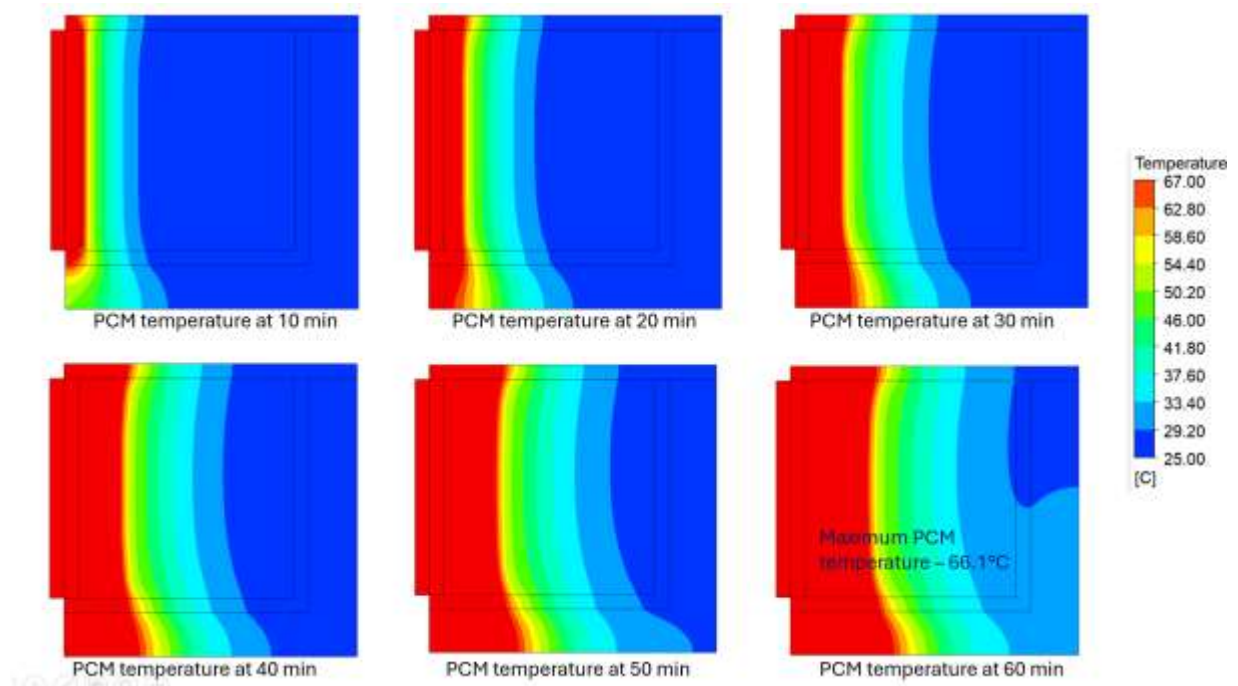


Figure 8. Temperature contours of PCM at midplane at 5W heater input for Heat sink A using OM55

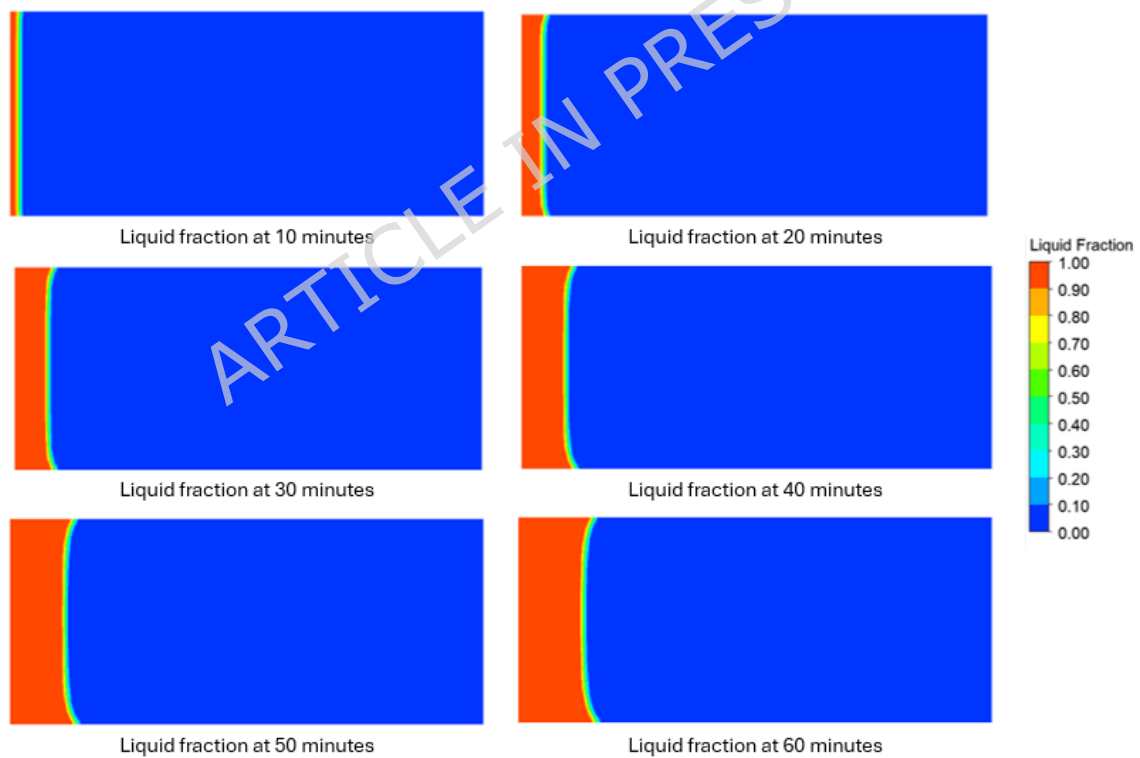


Figure 9. Variation of PCM liquid fraction over 1 h time duration for heater input of 5 W for Heat sink B using OM55

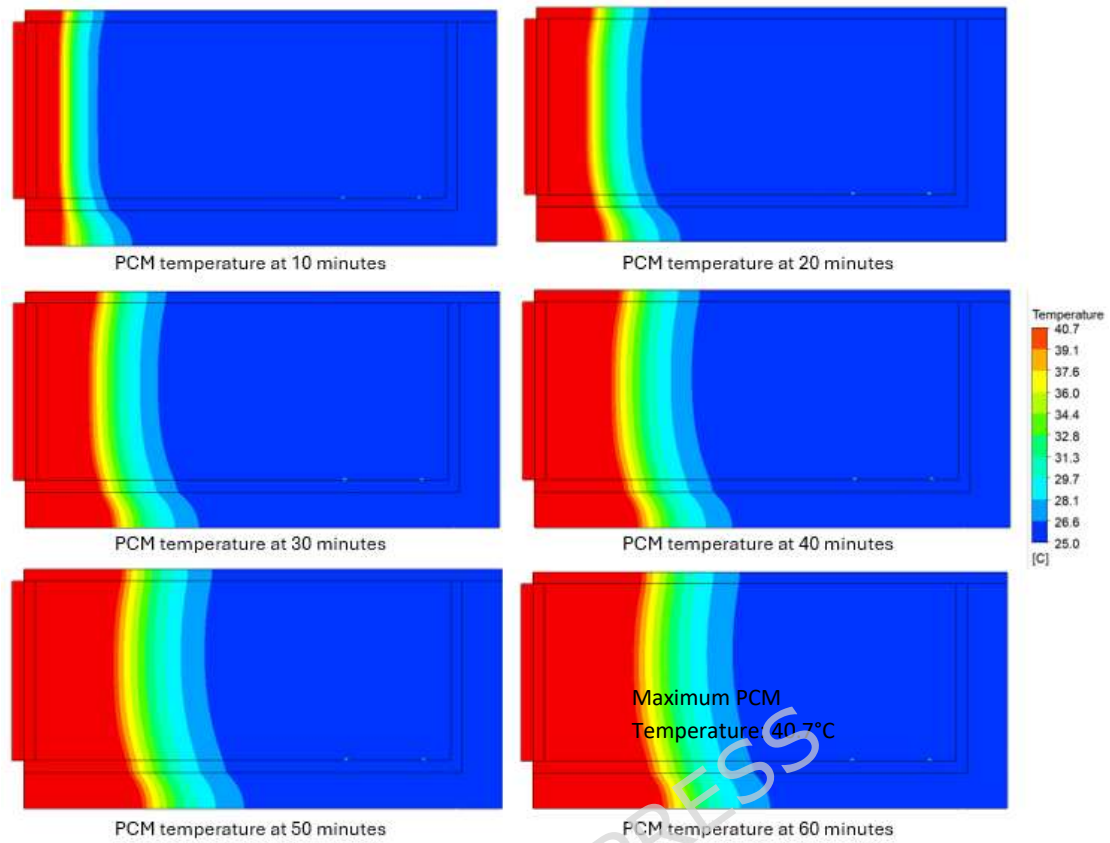


Figure 10. Temperature contours of PCM at midplane at 5W heater input for Heat sink B using OM55

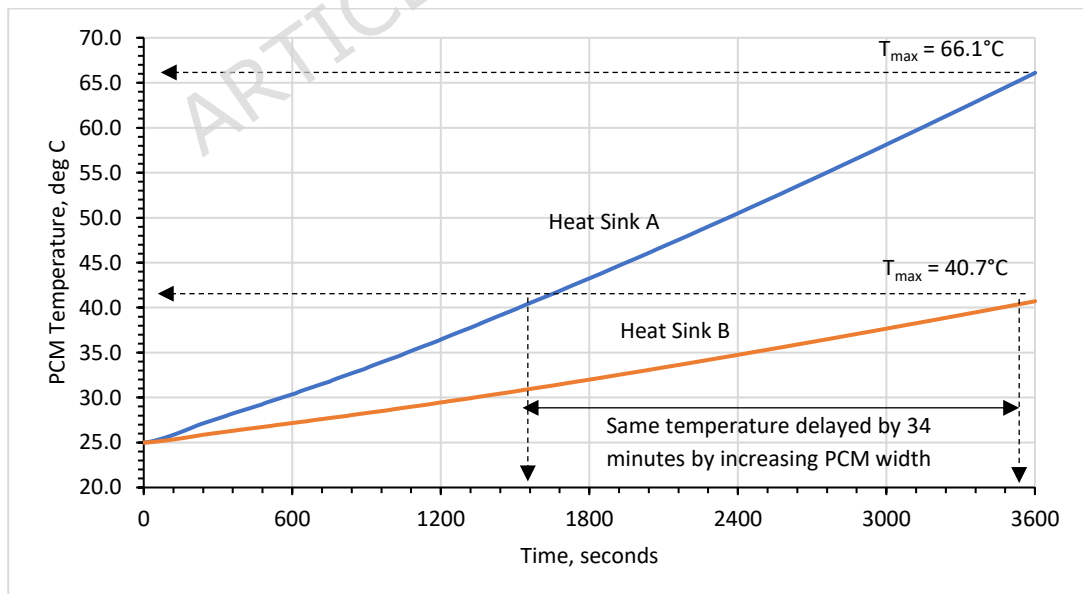


Figure 11. Effect of width of PCM on temperature profile for 5W heater input – Comparison between Heat sink A and Heat sink B

It may be noted that the PCM reached a maximum temperature of 66.1°C with heat sink A having a width of 60mm whereas for the heat sink B having a width of 120mm the temperature reduced to 40.7°C. This indicates that with increase in the width of PCM the temperature rise in any electronic equipment can be contained with the present study showing a substantial reduction of 38.4%. It can also be observed from Figure 11 that increase in width of PCM subsequently resulted in achieving the same temperature after a time period of 34 minutes confirming the effectiveness of increasing the heat sink width.

Having estimated the temperature rise with numerical studies, it is imperative to co-relate them with experimental work. The subsequent section compares the results of heat sink performance with phase change materials obtained experimentally.

3.2. Experimental Investigation on Effect of PCM on heat sink:

The following experimental procedure was used to investigate the performance of the PCM heat sink:

1. The electronic device was powered on and the power level was adjusted to produce the desired heat load.
2. The temperature sensors were monitored to record the temperature distribution of the electronic device and the PCM heat sink.
3. The experiment was continued until the PCM heat sink reached a steady state temperature.

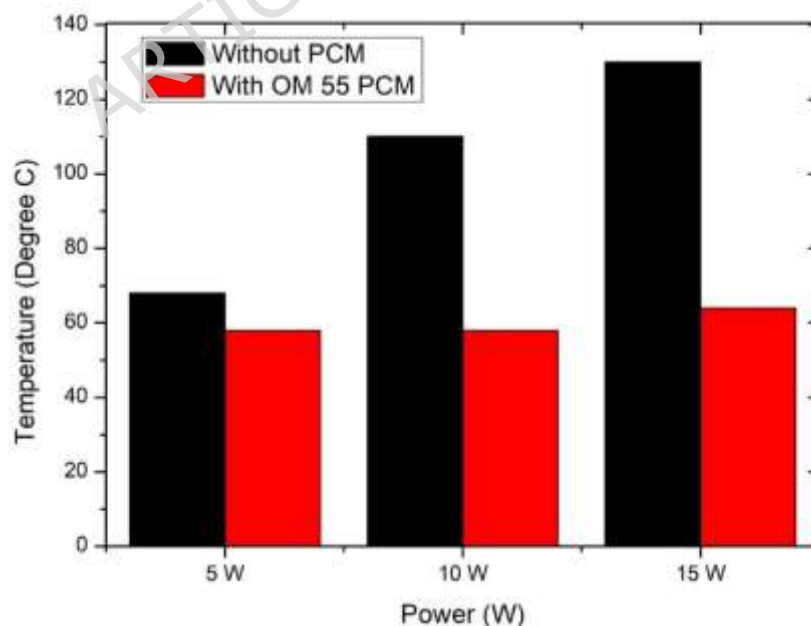


Figure 12. Comparison between PCM and without PCM heat sink temperatures for 5W, 10W, 15W power ratings.

The temperature measurements for PCM filled heat sink with and without PCM were taken for 60 minutes at heat power ratings of 5W, 10 W and 15 W. Fig. 12 compares the heat sink temperatures for the PCM-filled (OM55) and PCM-free heat sinks during the thermal regulation period. In case of heat sink without PCM, there was a remarkable rise in temperature to about 70°C, 100°C, and 120° C for power ratings of 5 W, 10 W, and 15 W, respectively. The rapid rise in temperature may affect the performance of electronic devices.

When compared to heat sink without PCM, the temperature of the heat sink with PCM decreases by 10.8°C, 50.1°C, and 69.8°C respectively at 5, 10, and 15 W. The temperature drop (achieved using PCM) increases with increasing power rating. The results also show an inverse relationship between the heater output and the thermal regulation period for a given quantity of PCM.

3.2.1 Effect of PCM melting temperature

In order to investigate the impact of PCM melting temperature on the heat sink thermal regulation, two types of PCMs are used. Heat sink A was utilized and tested for one hour. The PCMs that were tested were OM42 and OM55, and their respective melting temperatures are 42 °C and 55 °C. It is possible to study the heat sink surface temperature (T1) for the OM42 and OM55 power ratings of 5 W, 10 W, and 15 W as shown in Fig. 13.

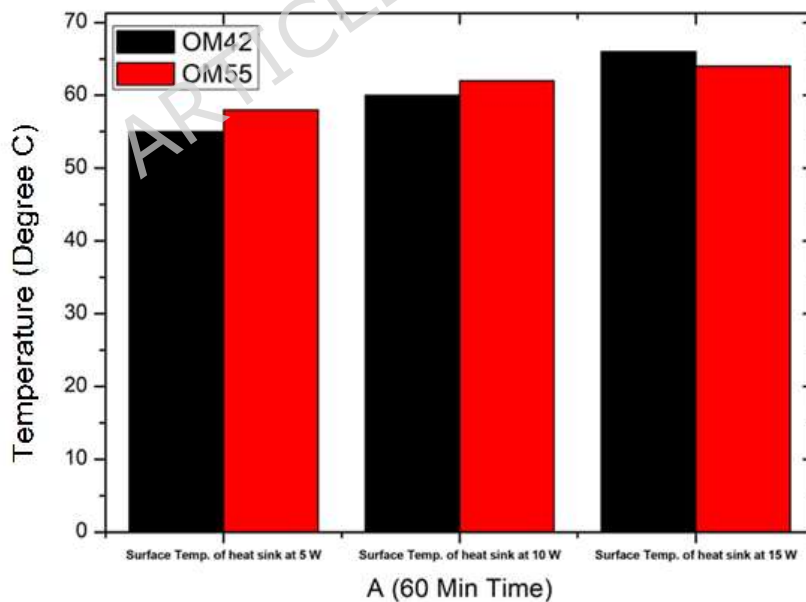


Figure 13. Surface temperature of heat sink A for OM55 and OM42 at (a) 5W, (b) 10W, (c) 15W power rating

The findings indicate that OM42 heat sink thermal regulation temperature is lower than OM55. This is due to the fact that OM42 PCM begins to melt and absorb heat in a latent form before OM55 PCM does. OM42 has a shorter thermal regulation period than OM55. Therefore, if the charging time is less than 50 minutes; OM42 is suitable to lower the heat sink maximum temperature for a 15 W power rating. A number of differences emerge when OM42 and OM55 heatsink types are compared based on the temperature increases they experience during an hour-long charge at various power ratings. At a power rating of 5W, both heatsink types exhibit no increase in temperature from their initial values, suggesting an effective cooling capability at this lower power input.

However, with increased power rating, OM42 consistently demonstrates a higher percentage increase in temperature compared to OM55. Specifically, at 10W, OM42 experiences about a 9.09% temperature rise, whereas OM55 registers a slightly lower increase of approximately 6.89%. This trend continues at 15W, where OM42 shows a more substantial rise of about 20% in temperature, while OM55's temperature elevation remains comparatively lower at approximately 10.34%. Consequently, at higher power ratings, OM42 appears to have a more pronounced temperature response, indicating a potentially lower thermal efficiency or lesser heat dissipation capability compared to OM55 under these conditions. Also, OM55 is able to reduce slightly the heat sink temperature for 10W.

3.2.2. *Effect of PCM thickness*

In order to understand the effect of PCM mass (thickness) on the temperature regulation period, experiments were conducted with heat sink A and B both. Fig. 14 shows the melting profile for OM55 at thickness of 60 mm and 120 mm for 5W power rating recorded at 20 min time intervals.

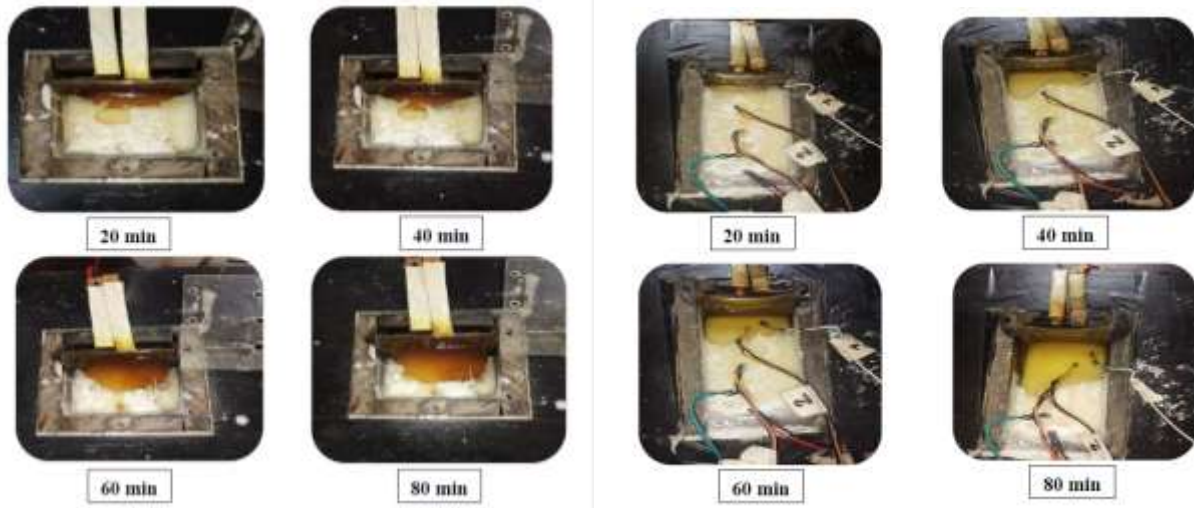


Fig. 14(a) Heat sink A at 5 W with thickness of 60mm(b) Heat B sink at 5 W with thickness of 120 mm

Observation of the pictorial representation in Figure 12 reveals that since the PCM closest to the heat source melts more quickly, it causes the liquid to push upward and the solid PCM to shift downward, giving the PCM its curved shape. Further, it was found that the period of thermal regulation was extended by 35 to 50 minutes. This closely matches the melting profiles shown in Figure 5 to 8 from the numerical simulation and also validates the observation of rise in regulation period of 34 minutes observed from numerical simulation.

To summarize, the thermal regulation mechanism is primarily governed by latent heat absorption during phase change, which delays temperature rise in the heat sink. Initially, heat transfer is conduction-dominated, causing the PCM temperature to rise until it reaches its melting point. Once melting begins, the PCM absorbs a significant amount of heat at nearly constant temperature, thereby stabilizing the system temperature.

- Effect of PCM melting temperature:

PCM with a lower melting temperature (OM42) starts melting earlier, resulting in quicker temperature control but a shorter regulation period due to faster depletion of latent heat. In contrast, PCM with a higher melting temperature (OM55) delays the onset of melting but sustains heat absorption over a longer duration, thereby extending the thermal regulation period, especially at higher heat inputs.

- Effect of PCM thickness (mass):

Increasing PCM thickness increases the total latent heat storage capacity. A larger PCM

volume requires more time to completely melt, which prolongs the duration over which heat is absorbed and thus extends the thermal regulation period. However, thickness has limited influence on peak temperature reduction because the initial heat transfer near the heater remains conduction-controlled.

Additionally, as melting progresses, natural convection within the liquid PCM enhances heat transfer, particularly near the boundaries, further influencing the melting front evolution and temperature distribution.

3.3. Comparison between experimental and numerical results

Table 6 compares the experimentally measured and numerically anticipated temperatures for various heat sink layouts utilizing OM55 PCM at a constant power input of 5 watts. For Heat Sink A, the numerical model predicts a temperature of 66.1°C vs the observed measurement of 58°C, resulting in a 13.7% error. This degree of agreement suggests that the simulation and experimental results are reasonably consistent.

Table 6. Comparison between experimental and numerical results

Heat Sink Configuration	PCM	Power rating, W	Experimental temperature, °C	Numerical temperature, °C	Error percentage
Heat Sink A	OM 55	5	58	66.1	13.7
Heat Sink B	OM 55	5	36	40.7	13.05

The deviation from the numerical results is of the order of 13 % as observed from Table 6. Uncertainties in PCM thermophysical properties, simplifications in phase change modelling especially the solid and liquid phase thermal conductivity, disregard for contact resistances, and experimental uncertainties like sensor accuracy are responsible for the observed 10–15% discrepancy between experimental and numerical results. These variations are in line with published research on PCM-based thermal systems (Mazo et al., 2015).

4. Conclusion

Phase change materials are a promising passive solution that can be utilized to absorb heat generated by electrical equipment such as computers or smartphones thereby providing essential cushioning against overheating and maintaining the operating temperature of these devices in a desirable

range. This study compares the performance of a laboratory designed prototype of heat sink so as to test its efficacy in proving thermal stability. Two different designs of cuboidal heat sinks integrated with regulated heaters – one 60 x 60 x 60 mm and 60 x 120 x 60 mm were utilized to study different factors like the thickness of the PCM and its melting temperature, different power ratings on the stable thermal regulation period. Numerical simulations were carried out to predict the advantage in establishing a stable thermal regulation period.

In order to understand efficiency of thermal regulation, the designed prototype heat sinks were subjected to different power ratings ranging from 5W to 15W for the following cases (i) with PCM to one without PCM (ii) Effect of melting temperature (iii) Effect of thickness of the PCM. The significant highlights from the experimental investigation work are as under:

- PCM integration can maintain temperature in the range of $\pm 5^{\circ}\text{C}$
- The maximum temperature can be lowered by 10 to 70 $^{\circ}\text{C}$ when PCMs are embedded in electronic devices.
- Increasing the melting temperature of the PCM from 42°C (OM42) to 55°C (OM55) extended the stable temperature time from 30 to 70 minutes (at power rating of 15 W)
- Maximum temperature reached is less sensitive to PCM thickness (PCM mass)
- Increasing PCM thickness by 2 times can raise thermal regulation period by about 50 minutes and 35 minutes for devices with power ratings of 5 watts and 15 watts respectively.

The findings of this study demonstrate the significant potential of PCMs in reducing heating issues in electronic equipment, that can enhance performance, longevity, and dependability. Future research directions may explore optimizing PCM formulations, scaling up applications, and integrating PCMs into diverse electronic cooling systems to further advance their practical implementation and impact in the field of thermal management. This study encourages continued exploration and adoption of PCM-based solutions to address the evolving challenges of electronic device heat dissipation.

Competing Interest:

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statements:

The authors confirm that the data supporting the findings of this study are available within the article.

Funding Declaration

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